# CRITICAL RESEARCH REVIEW ARTICLE

# [THE ROLE OF DIGITAL FORENSICS IN CYBERCRIME INVESTIGATIONS]


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### Abstract

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The abstract should include:

- Background/context of the selected research topic
- Purpose of the review
- General approach used to select and analyse the research articles
- Major findings from the reviewed literature
- Key research gaps identified
- Overall conclusion

**Recommended length: 150–250 words.**



**Keywords:** Digital Forensics; Cybersecurity; Investigation; Multimedia forensics; Cybercrime; Biometric; Data privacy

# Contents

# 1 Introduction

## 1.1 Background

As digital technology has experienced rapid evolution and expansion throughout the years, it has slowly integrated itself into most aspects of our own lives, from communications to our governance. Naturally, this boon that provides us convenience and connectivity will bring upon malicious actors that will misuse them to commit cybercrimes, ranging from small scaled crimes like identity theft and cybertheft of financial and card payment data to much larger scale ones such as cyberespionage and ransomware attacks.

As such, traditional methods of investigation have been proven to be ineffective in combating and solving these crimes. Hence, digital forensics as a discipline has seen a rise within the fields of cybersecurity and law enforcement to ensure that the malicious actors will have their deserved justice dealt to them alongside learning their methods to better suppress the surge of cybercrime.

## 1.2 Problem Statement

Almost every part of daily life now runs through a computer or a phone, and crime has followed right along with it. Money gets stolen, systems get broken into, and all of it can happen from the other side of the world. Digital forensics — the work of digging through devices and networks for proof of what happened — has become central to how law enforcement operates. That said, the field is under real strain right now.

For a long time, the basic strategy was straightforward: keep a list of known bad signatures and block anything that matched. That worked fine when attackers were unsophisticated. They aren't anymore. Advanced Persistent Threats, for example, are designed to slip past defenses and remain undiscovered for lengthy periods of time, silently removing important data from a network without raising any alarms. The scope of the problem is difficult to overestimate – cybercrime is expected to cost the global economy 10.5 trillion dollars by 2025, yet discovery is sometimes excruciatingly slow: one research estimated that the average time to find a breach was 212 days.

Part of the difficulty stems from the fact that digital forensics is a collection of disparate fields. Network forensics works with traffic, mobile forensics with phones and tablets, and then there's cloud, database, IoT, and multimedia forensics , each with its own set of tools and standards. A case that involves more than one of these areas requires investigators to piece together methodologies that were never intended to function together, slowing things slowly and introducing opportunity for mistake .

Criminals aren't making this easier, either. Encryption scrambles communications. Steganography hides malicious code inside what looks like an ordinary image. Fileless virus operates totally in memory and leaves no trace when the system reboots. Some players are even using blockchain to hide their actions. Even when investigators uncover anything, it does not always make it to court – shoddy paperwork, uneven processes, and contradicting legislation across jurisdictions are major reasons solid evidence is tossed out.

Artificial intelligence is frequently presented as the solution. It's easy to see why: machine learning is genuinely good at finding patterns buried in massive datasets and flagging behavior that a human analyst might miss , and there's evidence that AI-powered tools outperform older methods in terms of detection rate and false positives . But AI brings its own baggage. These models are frequently black boxes, so it's not always clear why a system flagged what it flagged. They can be manipulated with carefully designed adversarial inputs. They require massive training datasets, which are not always accessible, as well as significant computational capacity to function.

Then there's the smart-device problem. Homes are full of smart speakers, locks, thermostats, TVs; factories run thousands of interconnected sensors. All of this results in a massive, jumbled volume of data in inconsistent forms, dispersed among devices, cloud servers, and local storage. Figuring out where the relevant evidence even lives, let alone extracting it and proving it hasn't been tampered with, is genuinely difficult in this kind of distributed environment.

The legal landscape doesn't help. Privacy laws vary greatly by country; some countries openly share evidence while others do not, and permissible investigation tactics range from one another. Investigators must operate within this patchwork while adhering to guidelines such as GDPR and CCPA, and a misstep here might torpedo a case completely.

Despite years of research in this area, there is still no cohesive method that tackles all of these issues simultaneously. Standardized testing techniques for forensic tools are sparse, multi-domain datasets are scarce, and technologies produced by various manufacturers frequently do not work together. Furthermore, new dangers such as deepfakes and adversarial assaults continue to emerge and necessitate their own solutions.

## 1.3 Objectives of the Review

This study aims to assess where digital forensics is today: what has been solved, what has not, and where the profession appears to be headed — with a focus on how AI and similar technologies might boost investigations and protect evidence integrity.

More specifically:

First, we look at current practice in six domains: network, mobile, database, IoT, cloud, and multimedia forensics, highlighting the key issues, the tools and methodologies used, and the largest gaps in each.

Second, we evaluate how well AI-based methods function in practice, focusing on deep learning, federated learning, and anomaly detection to identify where these approaches succeed and where they fail .

Third, we examine the problems provided by new threats such as APTs, encryption, anti-forensic tactics, steganography, and AI-generated media, as well as potential responses .

Fourth, we look at the legal and ethical frameworks that shape the field, such as how courts assess the admissibility of digital evidence, chain-of-custody requirements, privacy protections, and the challenges that come with cross-border investigations.

Fifth, we present a cross-domain framework that combines federated learning with context-aware policy enforcement, which is intended to function across hybrid cloud, IoT, and corporate contexts while addressing the constraints of current single-domain techniques.

Finally, we highlight areas for future research, including explainable AI, quantum-resistant encryption, blockchain-based evidence tracking, privacy-preserving threat intelligence sharing, and automated attack attribution.

### 1.4 Research Questions

**RQ1:** What are the predominant methodologies and tools used in cybercrime investigations?

**RQ2:** What are the implications of artificial intelligence use in digital forensics?

**RQ3:** What are the methodological limitations identified in the current studies on digital forensics?

**RQ4:** What areas of digital forensics warrant further research?

## 2 Review Methodology

Explain how the group conducted the literature review.

### 2.1 Literature Search Strategy

To ensure the analysis we conducted are within the scope of our research and provides comprehensive depth, it is imperative to employ necessary techniques and paper selection for proper evaluation.

Firstly, a simple protocol was followed when selecting papers such that it is sourced explicitly using from trusted sources such as IEEE Xplore, ScienceDirect, IAES International Journal, arXiv, NUML International Journal. These academic databases were chosen for their viability to be cited, their coverage and for their contributions to digital forensics in cybersecurity.

A combination of keywords were used as search terms such as "digital forensics," "cybersecurity," "investigation," "artificial intelligence in digital forensics." These words along with "AND" and "OR" boolean operators used to refine the search and identify available and relevant papers.

Once the search had narrowed, papers were selected based on an initial screening based on title relevance to our subject, and were viewed in full-text to skim through its content.

Following the compilation of papers, an inclusion and exclusion criteria were used to determine whether it would be included or discarded.

### 2.2 Inclusion and Exclusion Criteria

To ensure the selection of research articles are of high quality and relevant to our research, a systematic inclusion and exclusion criteria were followed strictly.

**Inclusion criteria:**

1. Relevance: The paper must continuously be relevant to the current subject and contains at least one aspect of digital forensics.

2. Publication Date: Published between 2021 and 2026 to ensure that the studies discussed are updated and reflect current issues.

3. Page Length: Its pages were a minimum of 6 pages in length to ensure wide depth coverage on the subject.

4. Accessibility: The full-text is available at all times to be viewed for proper analysis.
5. Complete Information: Bibliographic information are complete and readily available for proper citation and referencing.
6. Coverage: The coverage of its subject is diversified and allows for new research areas to develop.
7. Research Contribution: the studies must have its own original findings through experiments or have a concise discussion related to digital forensics.

**Exclusion criteria:**

1. Published before the year 2021.
2. Page length is less than 6 pages.
3. Full-text is not readily available to view.
4. Insufficient bibliographic information for proper referencing.
5. Its research methodology is not readily disclosed and transparent.
6. Its title relevance to our subject were determined out-of-scope.
7. In a language other than English.

## 2.3 Article Selection

Following the initial compilation of research articles through title relevance, papers were rejected based on its page length being less than 6 pages, its publishing before 2021, and its insufficient research, amongst others.

Once compiled, each paper was meticulously examined through abstract inspection and the content of the main body's key findings and points. Concluding the article selection, 12 papers were selected based on the viability of adhering to the inclusion and exclusion criteria and was taken for literature review and further analysis.

# 3 Literature Review

## 3.1 Theme 1: Standardised Practices and Advancements of Digital Forensics

Through the rampant evolution of cybercrime in its sophistication of execution alongside the current digital age allowing access for various methods to execute cybercrime, digital forensics as a field within computer science has gone through many advancements to ensure that cybercriminals will face their due justice.

To better evaluate Digital Forensic's role in investigating and solving cybercrime, it would be apt to first understand the standardised methods of which the practitioners of digital forensics utilise in order to obtain the proper digital evidence for the committed cybercrime, and then delve into the various advancements made towards improving digital forensics' own practices as a whole.

Table 2 showcases research articles which align with the current theme of Standardised Practices of Digital Forensics, alongside their methodologies of research.

Table 1: Research Articles for Standardised Practices of Digital Forensics and their Methodology

| Research Articles | Primary Objectives | Methodology |
|---|---|---|
| Ilyas et al., 2024 | To analyze the extent and legal strength of digital forensic evidence within the Indonesian legal system. | **Normative**: Viewing law as a system of norms. Utilises philosophical (Nature of digital forensics), conceptual (Test and analyse concepts), and case (study and analyze electronic crime cases) approaches. |
| Hassan Shah Khan, 2025 | To evaluate the effectiveness of current digital forensic strategies and the possibility of enhancing them | **Qualitative**: Comparative analysis of digital forensic tools alongside examination of high-profile cybercrime case studies. |
| Sharma, 2024 | To investigate the adherence to legal standards and reporting procedures in forensic investigations. | **Quantitative Content Analysis**: Literature review of 40 reports from a quasi-experiment |
| Arif et al., 2025 | To evaluate the challenges of digital forensics the existing methodologies face | **Systematic Literature Review (SLR)**: Analysis of peer-reviewed publications from 2019 to 2025 across prominent databases. |

### 3.1.1 Comparison of Research Objectives and Scope

The chosen research articles intends to explore the transformation of digital forensics from traditional reactive models to much more reactive and intelligent frameworks.

- **General Evolution and AI Integration** : Research by Vaddi et al., 2024 focuses on the evolution of digital forensics over the past five years, specifically aiming to provide a high-level overview of AI's potential to solve modern difficulties. Their objective is to guide developers in constructing intelligent, automated tools for effective crime detection across blockchain, cloud, and deep learning (DL) domains Vaddi et al., 2024.
- **Smart Environments and Forensic Readiness**: Alenezi, 2023 shifts the focus toward "smart environments" (homes, cities, and IoT). The primary objective is to survey recent advancements and challenges in retrieving and preserving evidence across a multitude of connected devices, while proposing potential solutions to overcome the lack of standardization in these environments (Alenezi, 2023).

Table 3 showcases research articles which align with the theme of the Advancements of Digital Forensics, alongside their methodologies of research.

Table 2: Research Articles for Advancements of Digital Forensics and their Methodology

| Feature | Vaddi et al., 2024 | Alenezi, 2023 |
|---|---|---|
| Research Type | Passive data gathering, literature and statistical analysis | Comprehensive survey |
| Temporal Focus | Patterns no older than 2018 | Recent advancements and possible challenges |
| Core Methodology | Evaluating long-standing and latest techniques alongside summarising sections to keywords | Examining strengths and limitations of state-of-the-art tools and identifying major challenges and difficulties. |

#### 3.1.2 Standardised Investigative Procedures and Workflows

A central theme across the literature is the necessity of a structured, multi-stage forensic process to ensure the integrity and admissibility of evidence. While terminology varies slightly, the core stages remain consistent.

#### 3.1.3 The Forensic Lifecycle Comparison

The Forensic Lifecycle Comparison Standardised practices generally adhere to a flow of identifying, securing, and interpreting data.

- **Ilyas et al., 2024** outlines a four stage model: Identification, Examination, Analysis and Reporting. The article emphasises the digital evidence obtained through the four stage model to be volatile and must be stored carefully for proper presentation of the evidence in a trial to ensure it is not defective in the eyes of the law.
- **Hassan Shah Khan, 2025** instead presents a six-stage model: Identification, Collection, Preservation, Examination, Analysis, and Reporting. Key difference in comparison to the four stage model being the Collection and Preservation stages to ensure the integrity of the collected information, through use of write blockers, regular integrity checks and storage in secure, tamper proof containers
- **Arif et al., 2025** instead suggests a smoother model, consisting of Identification, Preservation, Analysis, Documentation, and Presentation. Their focus is on a high-level theoretical metamodel designed to unify tasks across subdomains like IoT and Cloud forensics to reduce ”process redundancies.”

#### 3.1.4 Key Procedural Requirements

Specific principles of digital forensics are treated as standardised practice, notably those cited from the Association of Chief Police Officers (ACPO) Ilyas et al., 2024 :

- Non-interference: Digital data is prohibited to be changed before being brought to court.
- Competence: Anyone accessing digital evidence must show a degree of competence alongside understanding of the implications and the relevancy of their actions upon said evidence.
- Documentation: Steps taken on the storage media while its being analysed and investigated must be audited detailedly so that they are able to be replicated exactly with the same results by a third party.

### 3.1.5 Dataset, Tools, and Technology Comparison

To be able to acquire proper digital evidence, a multitude of tools will be required. One particular research article, particularly Hassan Shah Khan, 2025 manages to compare and contrast multiple various software and hardware solutions, all utilised in different ways both professionally and academically, which is properly displayed in Table 4.

Table 3: Forensic Tools and Use Cases

| **Category** | **Tools Identified** | **Strengths** | **Limitati** |
|---|---|---|---|
| Computer Forensics | EnCase, FTK (Forensic Toolkit) | Comprehensive file systems and metadata analysis | High tool costs, hi demand |
| Mobile Forensics | Cellebrite UFED, MSAB XRY, Oxygen Forensic Suite | Effective data extraction from locked/encrypted mobile devices | Remote-wiping media, device |
| Network Forensics | Wireshark, NetWitness, Snort, Tcpdump | Real-time traffic monitoring and packet-level analysis | High expertise re steep learning curv |
| Cloud/IoT | Magnet AXIOM, IoT Inspector | Specialized for multi-tenant environments and smart device vulnerabilities | Jurisdictional, p protocol and ethic |

### 3.1.6 Comparative Analysis of Technical Frameworks and Tools and their Advancements

The articles highlights a transition toward automated, decentralized, and layered forensic architectures.

1. Artificial Intelligence and Machine Learning Both articles agree that AI and ML are essential for managing the volume and complexity of data generated in modern environments.

    - **Architectural Components**: Vaddi et al., 2024 propose a hybrid model consisting of a User Interface (for crime type selection and data entry), a Feature Extraction Module (using OCR and voice recognition), and a Crime Matching Engine. This engine utilizes multi-linear perceptrons, K-nearest neighbor (KNN), and neural networks Vaddi et al., 2024.
    - **Functional Applications**: Alenezi, 2023 categorizes AI/ML applications into image/audio analysis (facial/voice recognition), text analysis (sentiment and keyword identification in chat logs), and behavioral analysis (identifying suspicious patterns in user activity).

2. Cloud Forensics

    - **Layered Frameworks**: Vaddi et al., 2024 describes a cloud framework separated into four distinct layers: abstract, information gathering, data storage, and information fusion. This framework utilizes a "voting-k-means" approach for evidence separation Vaddi et al., 2024.
    - **Investigation Focus**: Alenezi, 2023 emphasizes the analysis of virtual machine images and the reconstruction of user activity within cloud-based services, highlighting the need for cloud forensic readiness to address data ownership and privacy (Alenezi, 2023).

3. Blockchain and IoT Integrity

    - **Chain of Custody**: Vaddi et al., 2024 present a forensic-chain framework for IoT. This utilizes a distributed ledger to record analysis events (Technical Evidence or TEs), ensuring immutability through Merkle trees and simplified Proof of Work (PoW) algorithms Vaddi et al., 2024.
    - **Evidence Management**: The blockchain framework uses smart contracts to automatically log every evidence transfer, including timestamps, permission levels, and recipient addresses Vaddi et al., 2024.

4. Network and Memory Forensics Alenezi, 2023 provides a detailed breakdown of network tools not extensively covered in the AI-centric Vaddi et al., 2024 study:
    - Packet Capture: Use of Wireshark, Snort, and tcpdump to identify suspicious traffic flows (Alenezi, 2023).
    - **Flow Analysis**: NetFlow and sFlow for detecting DoS and DDoS attacks.
    - **Memory Forensics**: Analysis of computer memory to uncover malware and network intrusions in IoT and mobile devices (Alenezi, 2023).

#### 3.1.7 Experimental Setup and Data Sources

The research articles utilizes diverse data points to validate their own discovery and findings:

- Ilyas et al., 2024: Used data from the Indonesian National Police Headquarters (e-MP), analyzing 8,831 cybercrime cases prosecuted in 2022.
- Sharma, 2024: Conducted content analysis on 40 reports generated via quasi-experiments to assess "Strength of Support" (SoS) categories.
- Hassan Shah Khan, 2025: Analyzed high-profile case studies including the 2014 Sony Pictures Hack, the 2021 Colonial Pipeline Ransomware Attack, and the 2018 Cambridge Analytical Data Scandal

In terms of the research articles focused on the advancements of digital forensics however prioritises simulation and pattern recognition over specific, localized datasets.

- **Data Sources**: Vaddi et al., 2024 emphasize the importance of diverse internet resources and training datasets for more accurate and precise findings. They specifically mention the use of unstructured crime report narratives and past records from external agencies Vaddi et al., 2024.
- **Visual Data**: Deep learning research focuses on forensic video analysis, specifically addressing low-quality CCTV footage. Techniques are devised to maximize the extraction of evidentiary elements through video enhancement Vaddi et al., 2024.
- **Heterogeneity**: Alenezi, 2023 notes that the dataset in smart environments is inherently heterogeneous, consisting of data from sensors, social media, and location-based services, which complicates correlation and reconstruction (Alenezi, 2023).

### 3.1.8 Results

The results of the research articles does reveals a discrepancy between theoretical standards and practical application:

- Limitations of Reports: Sharma, 2024 found that 82.5% of digital forensic studies were "Vague" in their methodology, and 0% explicitly asserted "Reliability" and "Validity" in a way consistent with other forensic sciences. This suggests a significant gap in academic and professional reporting standards.
- Legal Admissibility: Ilyas et al., 2024 concluded that while digital forensics is critical for finding "material truth," its success in Indonesia is hampered by a limited number of experts and internal police regulations that do not fully support the complexity of contemporary cybercrime.
- Automation and AI: Arif et al. (2025) and Hassan Shah Khan, 2025 both noted a rising trend in the use of AI and Machine Learning to automate evidence classification and anomaly detection, which helps in managing the "explosion of technology" and data volumes.

On the other hand, Table 5 showcases the resultant findings in regards to the possible advancements in terms of utilisation of Artificial Intelligence and Machine Learning:

Table 4: Comparative Results and Findings for AI and ML

| **Category** | **Vaddi et al., 2024** | **Alenezi, 2023** |
|---|---|---|
| Detection Accuracy | High efficacy in predicting illegal behavior and risk detection | AI/ML improves the speed and accuracy of analyzing large, complex datasets |
| Data Integrity | Effectively prevents evidence loss or corruption in insecure systems (Blockchain) | Forensic readiness policies reduce the time and cost associated with incident response |
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| Video/Media | Deep learning allows for extraction of evidence from low-quality footage | AI, through training is able to recognize specific suspects via voice or facial patterns |

### 3.1.9 Advantages and Limitations of Current Practices

The research is able to identify the benefits of standardised practices alongside the advancements of digital forensics while acknowledging significant barriers to universal implementation as well as the flaws of said advancement of digital forensics.

In regards to the Standardised Practices of Digital Forensics, the benefits and the flaws of its implementation are as follows:

- Advantages

- **Consistency**: Frameworks like the NIST guidelines provide a "consistent method of data retrieval" Hassan Shah Khan, 2025).

- **Legal Integrity**: Adherence to chain-of-custody protocols ensures that evidence remains "legally solid" and admissible in court Ilyas et al., 2024.
- **Efficiency**: Standardised tools like Magnet AXIOM allow for cross-platform recovery (Cloud, Mobile, Computer), reducing the need for disparate systems.

■ Limitations

- **Encryption and Obfuscation**: Advanced encryption methods act as a primary barrier, preventing access to data despite standard protocols Hassan Shah Khan, 2025).
- **Jurisdictional Complexity**: Cybercrime is inherently global, and the lack of a harmonized legal frameworks makes cross-border evidence sharing difficult (Arif et al., 2025; Sharma, 2024).
- **Rapid Obsolescence**: The speed of technological change often outpaces the development of forensic standards, particularly in the IoT and cloud sectors Hassan Shah Khan, 2025).
- **Vague Reports**: As identified by Sharma, 2024, a lack of specific methodological reporting hinders the ability of practitioners to replicate findings or rely on published research in a legal context.

In regards to the Advancements of Digital Forensics, the benefits and the flaws of its implementation are as follows:

■ Advantages

- **Automation**: Both studies highlight that AI and ML automate time-consuming manual processes, such as the visual scrutiny of video footage or the parsing of massive log files Alenezi, 2023.
- **Transparency and Security**: The use of blockchain provides decentralization, removing the vulnerability of a "central authority" that might be targeted by attackers Vaddi et al., 2024.
- **Proactive Readiness**: Forensic readiness planning allows organizations to collect evidence in a "forensically sound manner" before an incident occurs (Alenezi, 2023).

■ Limitations

- **Technical Hurdles**: Vaddi et al., 2024 acknowledge the impossibility of an error-free framework, noting the persistence of false positives. Alenezi, 2023 highlights the challenges posed by encryption and the lack of standardization in communication protocols across different devices.
- **Ethical and Legal Constraints**: Both sources point to significant challenges regarding data privacy. Alenezi, 2023 specifically notes the difficulty in balancing the collection of personal data against the need to prosecute cybercrimes.
- **Resource Intensity**: Implementing these advanced frameworks (e.g., establishing a digital forensic laboratory or maintaining a blockchain network) requires substantial material and human resources (Alenezi, 2023; Vaddi et al., 2024).

## 3.2 Theme 2: AI Augmented Digital Forensics

The rapid evolution of cybercrime has necessitated a shift from traditional manual forensic techniques to AI-augmented digital forensics. As digital evidence proliferates across cloud platforms, encrypted services, and IoT devices, the volume and velocity of data have exceeded human processing capacities.

Through our gathered research articles, they highlight the integration of Artificial Intelligence (AI) not as a replacement for human judgment, but as a critical instrument for enhancing the speed, accuracy, and depth of cybersecurity investigations. (Sanna et al., 2025; Zubair et al., 2025).

### 3.2.1 Comparative Analysis of Research Focus and Methodologies

The provided research context explores AI's role in digital forensics (DF) through different lenses, ranging from theoretical procedural frameworks to specific algorithmic testing and defense against sophisticated threats.

### 3.2.2 Similarities in Research Perspectives

Across the studies, several consensus begin to emerge:

- **The Scalability Crisis**: Researchers agree that traditional forensic methods is incapable of keeping up with the terabytes of logs and chaotic evidence produced during enterprise breaches Zubair et al., 2025.
- **Human-Centric AI**: AI is framed as an "investigative partner" or "digital assistant" rather than a "black box" solution. Human supervision is deemed essential to validate findings and address the legal requirement for interpretability in court (Sanna et al., 2025; Zubair et al., 2025).
- **The Adversarial Nature of AI**: There is a shared recognition in that cybercriminals are leveraging AI for offensive purposes as well, such as automated intrusion, phishing, and anti-forensics techniques like steganography (SN et al., 2025; Sanna et al., 2025).

### 3.2.3 Methodological Differences and Research Goals

The researchers employ distinct strategies to address the advancement of the field through utilisation of AI, as demonstrated in Table 6.

Table 5: Comparative Results and Findings for AI and ML

| Feature | Sanna et al., 2025 | Zubair et al., 2025 | SN et al., |
|---|---|---|---|
| Primary Focus | Procedural integration across the DF pipeline (4 Phase Procedure) | Technical performance of hybrid deep learning model | Defense against Persistent Threa |
| Research Methodology | Qualitative framework proposal and LLM case studies | Quantitative testing using CNN-BiLSTM on synthetic datasets | Comprehensive liter and APT case stu |
| Key Entities | General cybercrimes, LLMs (Gemini, Copilot, ChatGPT) | Ransomware, DDoS, and Insider Threats | State-sponsore Zero-day ex |
| Core Argument | AI should be applied at all phases of digital forensics (From collection to reporting) | Hybrid models (CNN+Bi-LSTM) with attention are superior to standard ML | APTs require a m defense (Zero Tr |

### 3.2.4 AI Applications in the Digital Forensic Pipeline

The integration of AI is analyzed relative to the standard four-phase forensic procedure defined by NIST: Collection, Examination, Analysis, and Reporting.

1. Collection and Preservation

   AI is able to facilitate the identification of digital evidence at crime scenes, through use of Computer Vision (CV) algorithms that can be deployed locally to assist first responders in identifying all relevant hardware (e.g., devices with volatile memory) to ensure all possible evidence has been recovered. AI also assists in terms of documenting the chain of custody by automatically recording and logging the device's conditions (Sanna et al., 2025).

2. Examination and Data Acquisition

   In the examination phase, AI serves as a digital assistant by guiding consultants through technical procedures, based on reviewing similar past cases. This is particularly useful for recovering data from corrupted memory sections or damaged disks. (Sanna et al., 2025).

3. Analysis and Evidence Interpretation

   This phase sees the most significant AI augmentation. Advanced models are used to:

   - Classify Multimedia: Automated identification of specific data types (e.g., images, audio, chats) enhances privacy by only analyzing relevant data while searching for evidence. Besides that, it helps reduces potential trauma gain from analysts viewing sensitive and explicit materials (Sanna et al., 2025).
   - Pattern Recognition: Deep learning models, specifically hybrid architectures like CNN-BiLSTM, excel at capturing both spatial dependencies in digital artifacts and temporal sequences in network traffic Zubair et al., 2025.
   - Anti-Forensics Detection: AI can be trained to detect hidden patterns in files used for steganography or to identify AI-generated "deepfake" content that mimics real individuals (Sanna et al., 2025).

4. Reporting and Legal Documentation Large Language Models (LLMs) and Natural Language Processing (NLP) are utilized to draft summaries and background technical definitions for legal parties. Future applications would include using NLP to review and score reports based on their understandability for non-technical stakeholders (Sanna et al., 2025).

### 3.2.5 Technical Performance and Model Robustness

Empirical testing suggests that advanced AI architectures significantly outperform traditional machine learning models in forensic tasks, as proven from Table 7 which uses Zubair et al., 2025 research of utilising a hybrid CNN-BiLSTM model with an attention mechanism and comparing it against baseline models such as Logistic Regression and Random Forest.

Table 6: Comparison of Model Metrics

| Model Type | Accuracy/F1-Score | Time-to-Discovery | False Positiv |
|---|---|---|---|
| Logistic Regression | 0.78 | High (12+ seconds) | 12.5% |
| Random Forest | 0.82 | Moderate ( 9 seconds) | 9.8% |
| Standard LSTM | 0.86 | Moderate ( 7 seconds) | 7.4% |
| CNN-BiLSTM + Attention | **0.93 - 0.96 (AUC)** | **Low ( 4 seconds)** | **3.1%** |

### 3.2.6 Strengths of the Hybrid Approach

The CNN-BiLSTM model's success is attributed to its **Attention Mechanism**, which assigns higher weight to important evidence segments. Ablation testing confirmed that removing attention decreased precision of the model, while removing metadata embeddings significantly decreased the model's recall Zubair et al., 2025.

### 3.2.7 Challenges and Contradictory Findings

While AI offers substantial benefits, the article accurately identifies a few critical limitations and areas of friction within the implementation:

- The LLM Steganography Paradox Sanna et al., 2025 conducted a case study on popular LLMs (Gemini, Copilot, ChatGPT) regarding steganography.
  - Failure in Creation: None of the LLMs could successfully generate a working PNG image with steganographed content.
  - Success in Scripting: Conversely, the LLMs were highly proficient at generating functional Python code for encoding and decoding messages using Least Significant Bit (LSB) techniques.
  - Insight: This suggests LLMs are currently more useful as an offensive or defensive coding assistants rather than as direct content generators for anti-forensics.
- Attribution Complexity in APTs APTs (Advanced Persistent Threats) present a unique challenge due to their stealthy and state-sponsored nature. Attribution remains difficult because malicious actors use custom malware and fileless techniques

that runs only within the system memory instead of any other vulnerable areas. While AI improves threat hunting, the **automated attribution** of these actors remains a primary research gap (SN et al., 2025).

- Data Bias and Environmental Variability A significant weakness in AI forensic tools is the risk of biased models. Models that are trained on Western corporate data may risk failure when applied to mobile-centric systems in developing countries. Ensuring fairness and accuracy requires diverse, globally-sourced training datasets/ Zubair et al., 2025.

### 3.2.8 Emerging Trends and Future Directions

While reading through the articles, a trend can be observed in the fact that digital forensics is moving towards more resilient, transparent and collaborative ecosystems.

- Explainable AI (XAI): There appears to be a necessity for XAI to be properly utilised to provide clear justifications for the AI's own decisions. This transparency is vital for establishing trust among investigators and ensuring that the obtained evidence is admissible in court (SN et al., 2025; Zubair et al., 2025).
- Blockchain Integration: Researchers are exploring blockchain as a much more secure method for identity management, tamper-proof logging of forensic procedures alongside providing a decentralized threat intelligence sharing platform. (SN et al., 2025).
- AI-vs-AI Feedback Loops: A proposed trend involves a continuous loop where offensive AI develops evasive techniques to better test defensive AI's responses. Human supervisors are required to monitor this interaction to better refine both of the AI's detection algorithms and analyst skills (Sanna et al., 2025).
- Federated Learning: This technique allows organizations to collaboratively train APT detection models without sharing sensitive private data, thereby preserving privacy while increasing the shared knowledge base (SN et al., 2025).

## 3.3 Theme 3: Cybercrime and the Detection of Digital Deceptions

### 3.3.1 Overview of the Cybercrime Landscape

The global cost of cybercrime is projected to reach $10.5 trillion by 2025 Osamor et al., 2026. This financial escalation is driven by the increasing sophistication of digital deception, primarily through two vectors: the democratization of high-fidelity synthetic media (deepfakes) and the refinement of traditional phishing via spoofed communications. Both methods exploit human trust and systemic vulnerabilities in standard authentication protocols.

### 3.3.2 Comparative Analysis of Deception Techniques

1. Deepfake Generation and Proliferation

   According to Singh and Dhumane, 2025, deepfakes utilize Generative Adversarial Networks (GANs) to create hyper-realistic images, audio, and video through techniques like face synthesis (e.g., StyleGAN), face morphing, and speech cloning (e.g., Wav2Lip). This could be explained through the fact that open-source tools like DeepFaceLab and Faceswap have lowered the barrier of entry, allowing non-technical actors to generate deceptive content Singh and Dhumane, 2025. No doubt the high prevalence of deepfakes like the Deepfake cases in India alone rose by 550%, with 90% of surveyed individuals in the UK expressing apprehension about the technology Singh and Dhumane, 2025.

2. Email Spoofing and Phishing

   Osamor et al., 2026 highlight that phishing remains a dominant threat, with a 1726% increase in phishing websites between 2011 and 2021. Cybercriminals are able to use tools like malicious IP addresses, forged email headers, and tracking pixels which are used to monitor the recipient's activity and location as an invasion of their privacy.

   Unlike deepfakes, which often rely on visual/auditory realism, email spoofing instead leverages social engineering triggers such as emergencies and sudden authority presence (e.g., "Microsoft account unusual sign-in activity") to bypass the victim's critical thinking Osamor et al., 2026.

### 3.3.3 Detection Methodologies

To better combat the cybercrimes, researchers have propose diverse approaches to identifying digital deceptions, ranging from automated AI models to manual forensic analysis, as can be seen in Table 8.

Table 7: Detection Techniques and Capabilities

| Technique | Application | Capabilities | Limitati |
|---|---|---|---|
| **CNN-Based Models** | Deepfakes (Image/Video) | High accuracy in spatial artifacts Singh and Dhumane, 2025 | Sensitive to comp resolutio |
| **RNN/LSTM Models** | Deepfakes (Image/Video) | Captures temporal inconsistencies and motion patterns Singh and Dhumane, 2025 | High computationa inferenc |
| **Header Analysis** | Email Spoofing | Identifies mismatches in SPF, DKIM, and DMARC Osamor et al., 2026 | Can be bypas sophisticated ob |
| **Biometric Defenses** | Identity Verification | Liveness detection (eye blinks, pupil dilation) and 3D depth sensing Singh and Dhumane, 2025 | Subject-dependent input qua |
| **Hybrid Approaches** | Multi-modal Detection | Combines human-in-the-loop verification with AI-based filtering Osamor et al., 2026 | Complex desi resource-h |

1. Similarities in Research Findings Both studies emphasize that **authentication failures** are primary indicators of fraud. In email forensics, this manifests as **spf=none** or **dkim=none** results Osamor et al., 2026. In deepfake detection, this involves identifying "spoofing" where synthetic data is presented as genuine biometric data Singh and Dhumane, 2025. Furthermore, both identify **demographic**

**and environmental biases** in training data (e.g., FFHQ and DFDC datasets) as a significant hurdle for universal detection Singh and Dhumane, 2025.

2. Differences and Contradictory Findings

   - Scalability: Singh and Dhumane, 2025 note that heavy AI pipelines (Transformers) are computationally expensive and struggle with real-time requirements (running at 18–22 fps). Conversely, Osamor et al., 2026 argue for the continued necessity of manual forensic analysis, asserting that automated tools often overlook subtle anomalies like a 2-second timestamp mismatch in email hops.
   - Human Factor: Osamor et al., 2026 emphasises human behavior as a **critical vulnerability**, whereas Singh and Dhumane, 2025 focuses more on the technical **arms race** between generative models and detection algorithms while arguing for the human element for the detector models.

#### 3.3.4 Strengths, Weaknesses, and Vulnerabilities

1. Technical Weaknesses

   - **Cross-Dataset Generalization**: Models trained on specific datasets (e.g., DFDC) often fail when tested against "in-the-wild" content. For example, the accuracy of the models can drop from ¿90% to 60% when facing user-generated content like compressed TikTok videos Singh and Dhumane, 2025.
   - **Adversarial Attacks**: Attackers can add **intentional perturbations** (noise) to forged media to deceive CNN-based classifiers into labeling them as authentic Singh and Dhumane, 2025.

2. Forensic Failures Real-world incidents demonstrate the stakes of these vulnerabilities:

   - **Political Turmoil**: The 2019 Ali Bongo incident in Gabon showed how a suspected deepfake (even if genuine) can trigger military takeover attempts due to public mistrust Singh and Dhumane, 2025.
   - **Financial Loss**: A 2019 voice deepfake attack against a UK-based CEO resulted in a $243,000 fraudulent wire transfer, illustrating the danger of voice cloning in high-stakes environments Singh and Dhumane, 2025.

#### 3.3.5 Emerging Trends and Policy Frameworks

- Research Trends While analysing the research articles for this review, certain trends tend to pop up. One in particular being the Explainable AI (XAI), which are models that provide reasonings alongside an interpretable output to support legal admissibility and public trust in their usage Singh and Dhumane, 2025. Another one would be the facilitation of Federated Learning, a privacy-preserving approach where models area able to be trained across devices without centralizing raw biometric data Singh and Dhumane, 2025. In addition, there seems to be talks of the implementation of continuous authentication: a system that monitor user behaviors (gait, keystroke rhythm) throughout a whole session rather than relying on a one-time login Singh and Dhumane, 2025, which could better ensure the user isn't a malicious actor that pretends to be another user.

- Global Regulatory Initiatives In terms of global efforts in the other hand, it appears that there are new legal frameworks being established, with the aim to categorize and mitigate AI risks. For one, the EU AI Act (2024) has categorised deepfakes as **high-risk**, requiring transparency measures such as labeling synthetic content Singh and Dhumane, 2025 to avoid misconception. As observable from NIST's Incident Response, Osamor et al., 2026 advocates for integrating the NIST Incident Response Lifecycle with current forensic investigations to standardize how organizations should handle spoofing attacks. In addition, a brand new strategy has appeared: **Provenance Tagging**, which uses cryptographic hashes to track the origin of digital media, is currently being supported by the Partnership on AI (2023) to identify tampered content Singh and Dhumane, 2025.

# 4 Comparative Analysis of Existing Studies

Provide a systematic comparison of the reviewed research articles.

A table such as the following is recommended.

Table 8: Comparative analysis of the reviewed research studies

| No. | Author & Year | Research Objective | Method/Technique | Dataset/Sample | Key Findings | Limitations |
|---|---|---|---|---|---|---|
| 1 | [Author, Year] | [Research objective] | [Method/Technique] | [Dataset/Sample] | [Key findings] | [Limitations] |
| 2 | [Author, Year] | [Research objective] | [Method/Technique] | [Dataset/Sample] | [Key findings] | [Limitations] |
| 3 | [Author, Year] | [Research objective] | [Method/Technique] | [Dataset/Sample] | [Key findings] | [Limitations] |
| 4 | [Author, Year] | [Research objective] | [Method/Technique] | [Dataset/Sample] | [Key findings] | [Limitations] |

# 5 Synthesis of Findings

This section is particularly important.

The group should answer:

## 5.1 What Do the Studies Agree On?

Identify findings or conclusions that are consistently reported across multiple studies.

## 5.2 What Do the Studies Disagree On?

Identify conflicting findings, different results, or different conclusions.

## 5.3 Which Approaches Appear to Be Most Effective?

Compare the approaches and explain why certain methods appear to perform better.

## 5.4 What Trends Can Be Observed?

Identify emerging trends in:

- Technologies

- Methodologies
- Datasets
- Algorithms
- Applications
- Research directions

The discussion should be based on evidence from the reviewed articles.

# 6 Research Gaps

Identify gaps that remain in the existing literature.

Possible research gaps include:

- Lack of suitable datasets
- Small sample sizes
- Limited geographical coverage
- Poor generalisation
- Limited experimental validation
- Lack of comparison between methods
- High computational requirements
- Lack of real-world testing
- Security/privacy concerns
- Lack of explainability
- Outdated technologies
- Limited integration with other technologies

For each identified gap, provide evidence from the reviewed literature.

# 7 Future Research Directions

Based on the identified research gaps, propose potential directions for future research.

For example:

1. Development of improved algorithms
2. Use of larger datasets
3. Integration of emerging technologies

4. Development of real-time systems
5. Improved accuracy/performance
6. Cross-domain applications
7. Improved security and privacy
8. Explainable AI approaches
9. Real-world deployment and validation

The proposed directions should logically follow from the research gaps identified in Section **??**.

# 8 Discussion

Provide an overall critical discussion of the literature.

The discussion should answer:

- What has the research community achieved so far?
- Which approaches are most promising?
- What are the major challenges?
- Are the current approaches sufficient?
- What limitations remain?
- How has the research area evolved?
- What important issues remain unresolved?

This section should demonstrate **critical thinking**, not merely summarisation.

# 9 Conclusion

Summarise the major findings of the review.

The conclusion should:

- Restate the purpose of the review
- Summarise the major findings
- Highlight important research gaps
- State the overall conclusion
- Briefly indicate future research opportunities

Do not introduce completely new information or references in the conclusion.

# References


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# APPENDIX A – INDIVIDUAL ARTICLE REVIEW

Each student should complete the following table for their **minimum 3 research articles** before the group combines the findings.

**Student:** [Name / Student ID]

Table 9: Individual article review

| **Article** | **Research Problem** | **Objective** | **Methodology** | **Dataset/Sample** | **Key Findings** | **Limitations** | **Resear Gap** |
|---|---|---|---|---|---|---|---|
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This appendix provides evidence of each student's individual contribution to the group review.

# APPENDIX B – ARTICLE MATRIX

The group should combine all individual article reviews into one matrix.

Table 10: Group article matrix of reviewed research studies

| Article | Student | Year | Method | Dataset | Key Results | Strength | Limitation | Research Gap |
|---|---|---|---|---|---|---|---|---|
| 1 | Student 1 | | | | | | | |
| 2 | Student 1 | | | | | | | |
| 3 | Student 1 | | | | | | | |
| 4 | Student 2 | | | | | | | |
| 5 | Student 2 | | | | | | | |
| 6 | Student 2 | | | | | | | |
| ... | ... | | | | | | | |